Lab 6

Set Up Nexus Repository and Configure User Access

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2023-BSCS-44

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Task 1: Run Nexus Repository and Complete the Initial Setup

- **nexus_container_running.png**

```
irzahameed@ubuntu:~$ docker ps | grep nexus
fc21a960716c   sonatype/nexus3:latest          "/opt/sonatype/nexus…"   53 seconds ago   Up 52 seconds   0.0.0.0:8007->8
007/tcp, [::]:8007->8007/tcp, 0.0.0.0:8006->8081/tcp, [::]:8006->8081/tcp   nexus
irzahameed@ubuntu:~$
```

Task 2: Create a Role and User and Verify Access

- **nexus_user_access.png**

```
irzahameed@ubuntu: ~

irzahameed@ubuntu:~$ curl -i -u nexus http://localhost:8006/service/rest/v1/status
Enter host password for user 'nexus':
HTTP/1.1 200 OK
Server: Nexus/3.98.1-01 (COMMUNITY)
Vary: Accept-Encoding
X-Content-Type-Options: nosniff
Content-Security-Policy: default-src http: data: blob: 'unsafe-inline'; script-src http: 'unsafe-inline' 'unsafe-eval'
Content-Length: 0
```

Task 3: Build and Publish the Gradle Application to Nexus

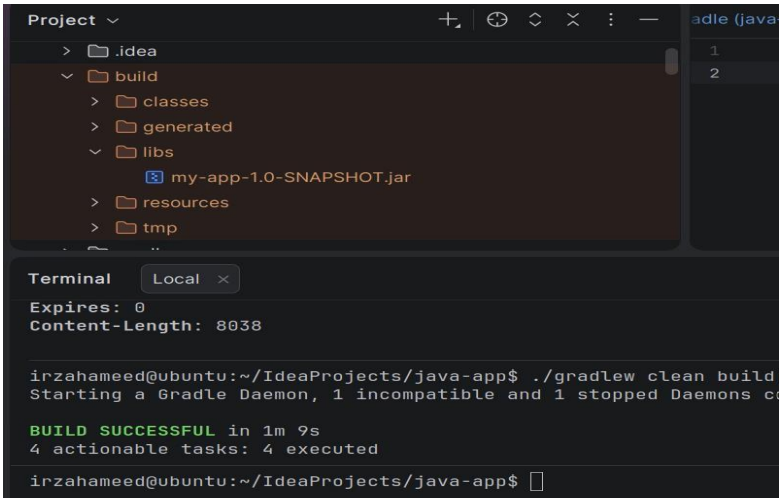
- **gradle_nexus_configuration.png**

README.md | build.gradle (java-app) | settings.gradle (java-app) | gradle.properties

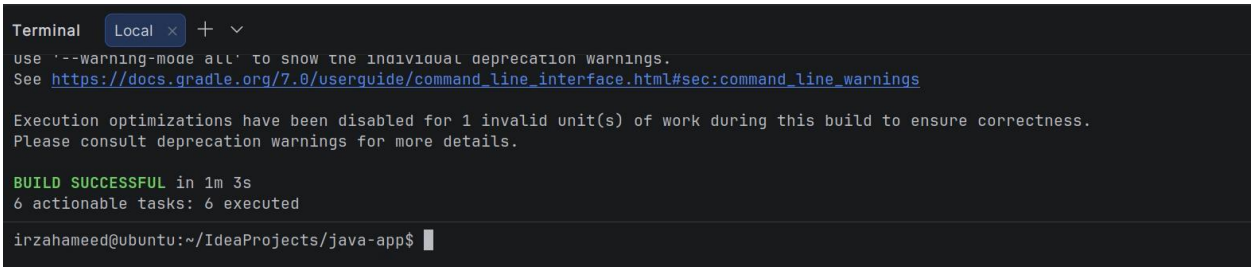
```
group 'com.example'
version '1.0-SNAPSHOT'

publishing {
    publications {
        maven(MavenPublication) {
            artifact("build/libs/my-app-$version" + ".jar") {
                extension "jar"
            }
        }
    }
    repositories {
        maven {
```

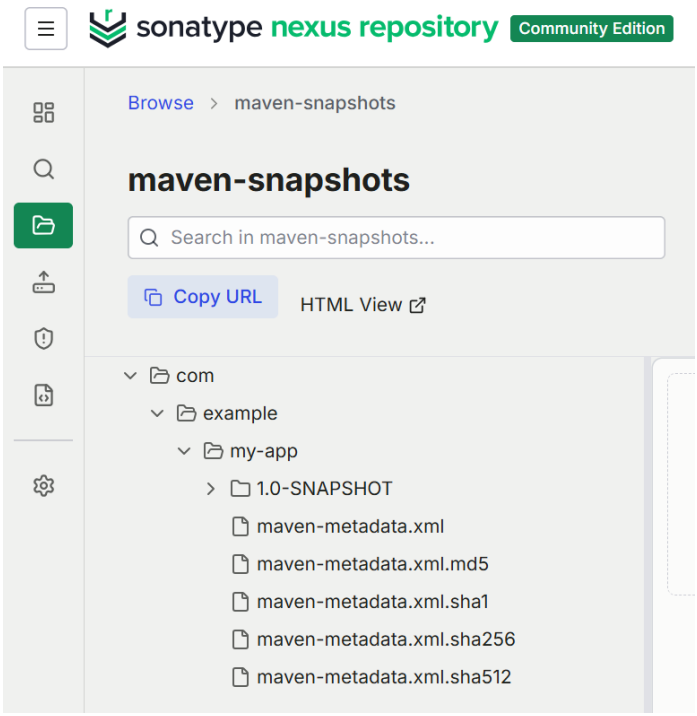
- **gradle_nexus_build.png**



- **gradle_nexus_publish.png**

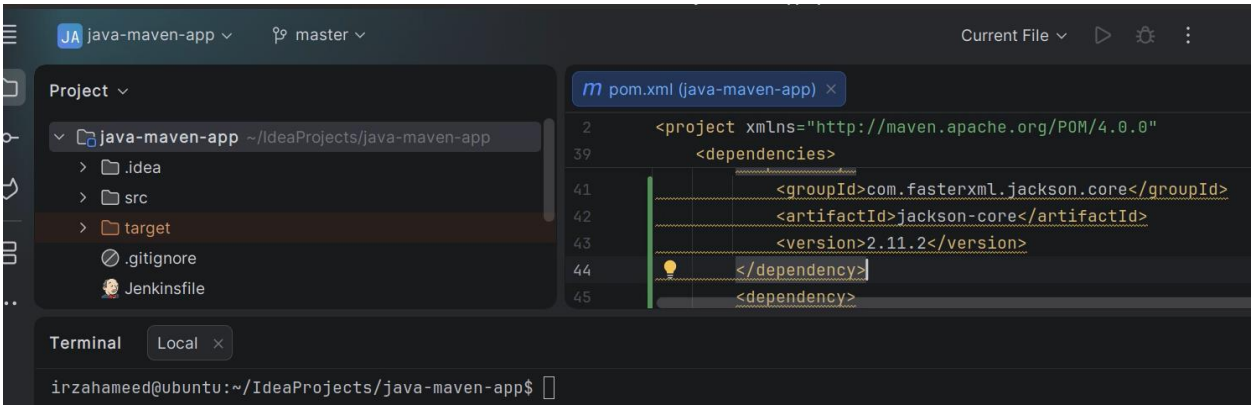


- **nexus_gradle_artifact.png**

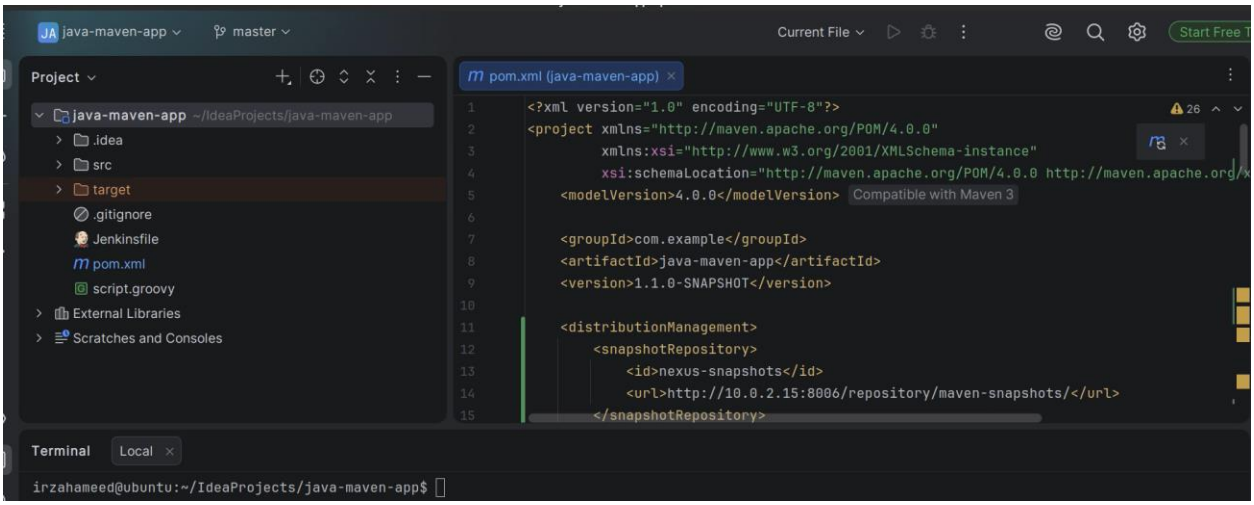


Task 4: Publish the Maven Application to Nexus

- **maven_nexus_project.png**



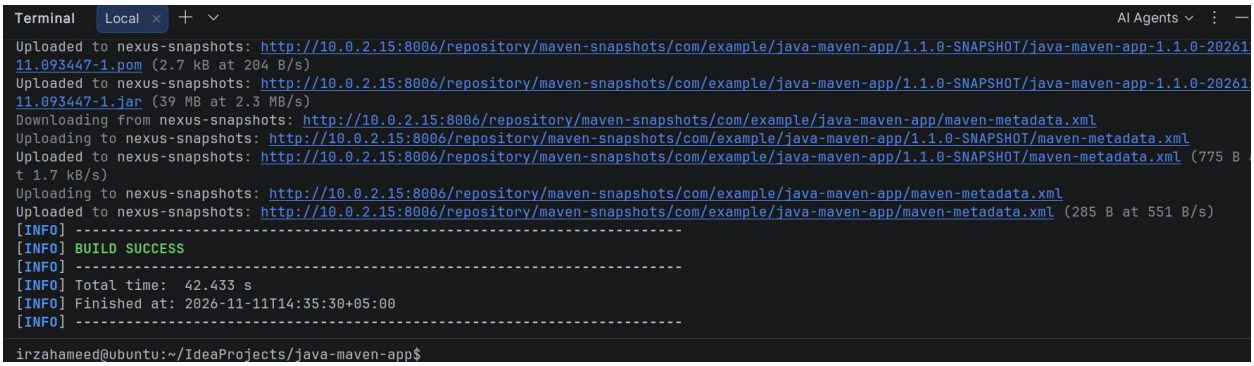
- **maven_distribution_management.png**



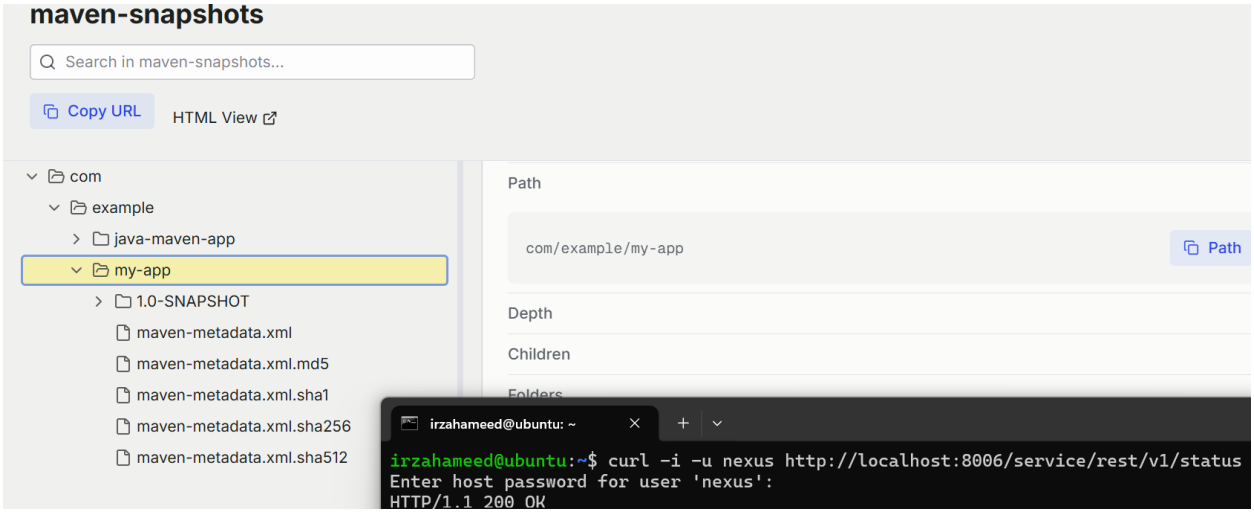
- **maven_package_nexus.png**



- **maven_deploy_nexus.png**

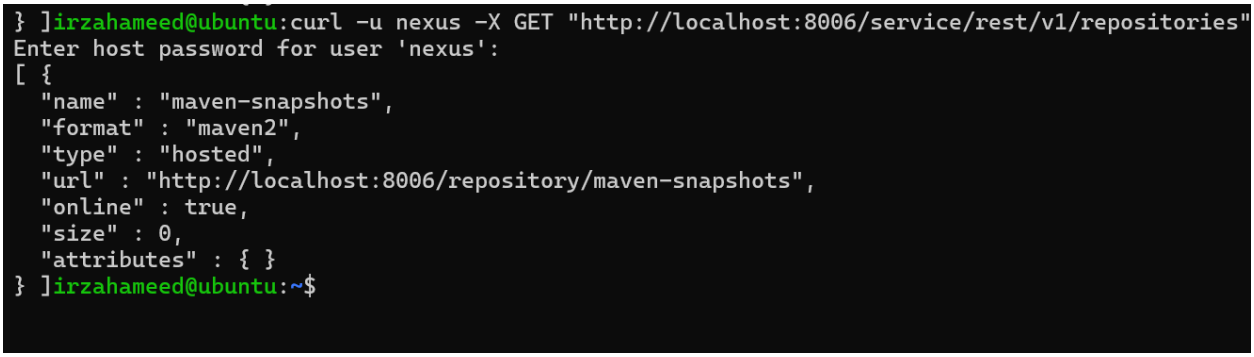


- **nexus_maven_artifact.png**

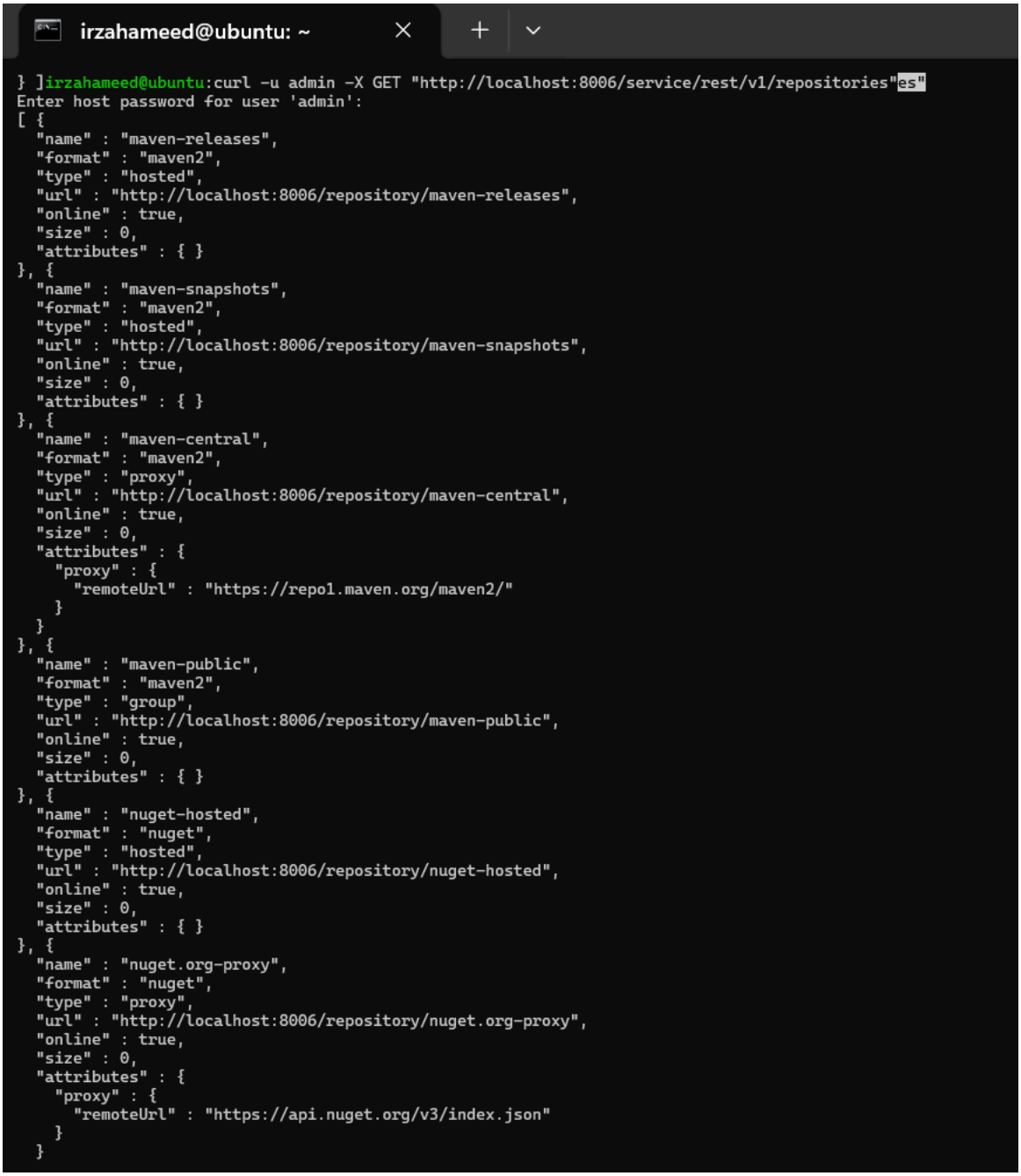


Task 5: Inspect Nexus Repositories and Components Using the REST API

- **nexus_repositories_user.png**



- **nexus_repositories_admin.png**



- **nexus_components_list.png**

```
irzahameed@ubuntu: ~
irzahameed@ubuntu:~$ curl -u nexus -X GET "http://localhost:8006/service/rest/v1/components?repository=maven-snapshots"
Enter host password for user 'nexus':
{
  "items" : [ {
    "id" : "bWF2ZW4tc25hcHNob3RzOjRmMWJiY2Rk",
    "repository" : "maven-snapshots",
    "format" : "maven2",
    "group" : "com.example",
    "name" : "my-app",
    "version" : "1.0-20261111.090810-1",
    "assets" : [ {
      "downloadUrl" : "http://localhost:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261111.090810-1.jar",
      "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261111.090810-1.jar",
      "id" : "bWF2ZW4tc25hcHNob3RzOmRhYTY2ZDJl",
      "repository" : "maven-snapshots",
      "format" : "maven2",
      "checksum" : {
        "sha1" : "b8087a14fc97b47220ab3afd65935ef8679fc8e0",
        "sha512" : "63920a3cc45801a836253c316872ce8efd50643f52efb3f0f28c7dc095bceece0fc6536bcd703ffceda944d6141577d7838690264 4c9fb36e5003c4167fa42b1",
        "md5" : "ab30d57836df58d2bd0c205202ce8f51",
        "sha256" : "bb58a6f2f6cc2f69ba01fe85f6e8d0216ed69e368c64a0a42a2b75f3dfe8d037"
      },
      "contentType" : "application/java-archive",
      "lastModified" : "2026-11-11T09:09:02.888+00:00",
      "lastDownloaded" : null,
      "uploader" : null,
      "uploaderIp" : null,
      "fileSize" : 39682480,
      "blobCreated" : "2026-11-11T09:09:02.895+00:00",
      "blobStoreName" : "default",
      "blobUpdated" : "2026-11-11T09:09:02.888+00:00",
      "blobRef" : "default@2327556d-ec02-4a0d-9c1f-d7c30ec2fde2@2026-11-11T09:09",
      "lastVerified" : null,
      "registryUrl" : null,
      "maven2" : {
        "extension" : "jar",
        "groupId" : "com.example",
        "artifactId" : "my-app",
        "version" : "1.0-20261111.090810-1"
      }
    }, {
      "downloadUrl" : "http://localhost:8006/repository/maven-snapshots/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261111.090810-1.jar.sha1",
      "path" : "/com/example/my-app/1.0-SNAPSHOT/my-app-1.0-20261111.090810-1.jar.sha1",
      "id" : "bWF2ZW4tc25hcHNob3RzOjI5YzIyYTBi",
      "repository" : "maven-snapshots",
      "format" : "maven2",
      "checksum" : {
        "sha1" : "77cb8dfc6fd42956c471a93719d02c8f2af23f9c",
        "sha512" : "e1bd2d988f30a6cf78121974c5febc6e1608f717e1684d0a0926fb389c10a8fad8b6b56eb8a76ba5ff57dfb00ff18d2756badfec83fe0cccf5177151dc2b4880",
```

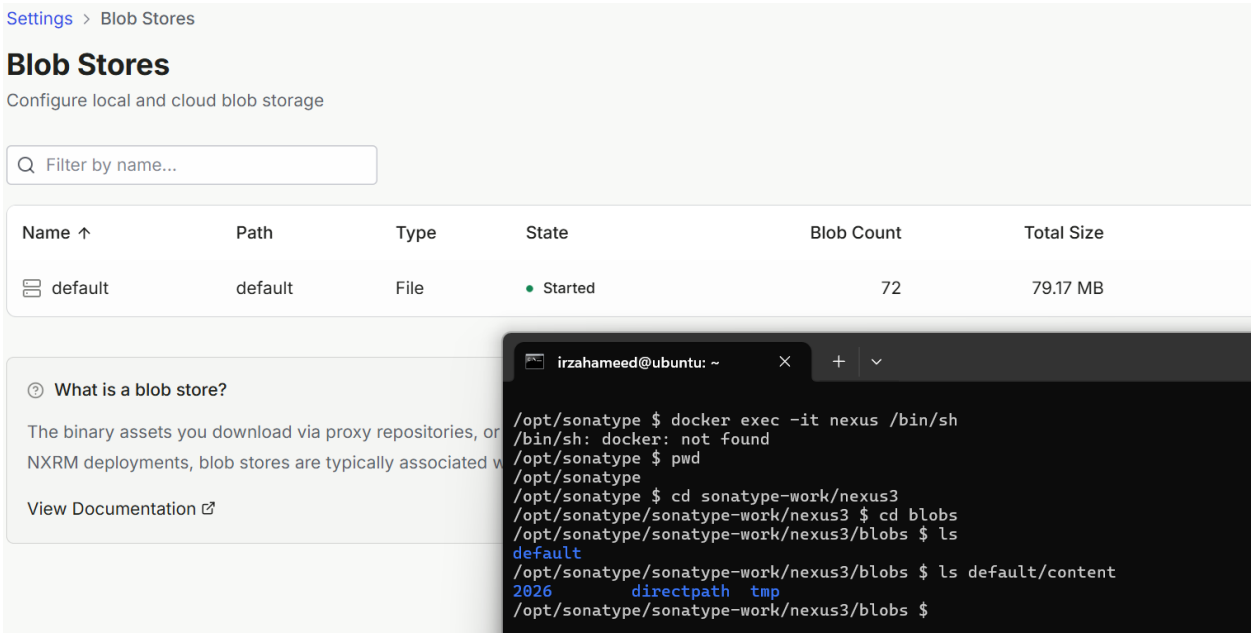
- **nexus_component_details.png**

```
irzahameed@ubuntu:~$ curl -u nexus -X GET "http://localhost:8006/service/rest/v1/components/<id>"
Enter host password for user 'nexus':
<html>
<head>
<meta http-equiv="Content-Type" content="text/html;charset=ISO-8859-1"/>
<title>Error 400 Bad Request</title>
</head>
<body>
<h2>HTTP ERROR 400 Bad Request</h2>
<table>
<tr><th>URI:</th><td>http://localhost:8006/service/rest/v1/components/&lt;id&gt;</td></tr>
<tr><th>STATUS:</th><td>400</td></tr>
<tr><th>MESSAGE:</th><td>Bad Request</td></tr>
</table>
<hr/><a href="https://jetty.org/">Powered by Jetty:// 12.1.13</a><hr/>

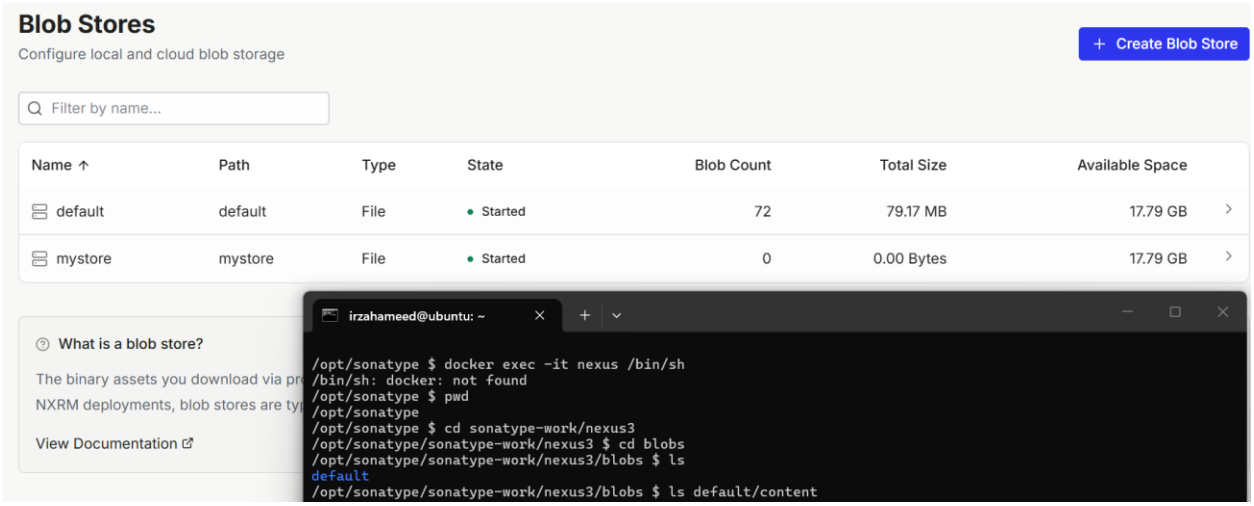
</body>
</html>
irzahameed@ubuntu:~$
```

Task 6: Create a Nexus Blob Store and Docker Hosted Repository

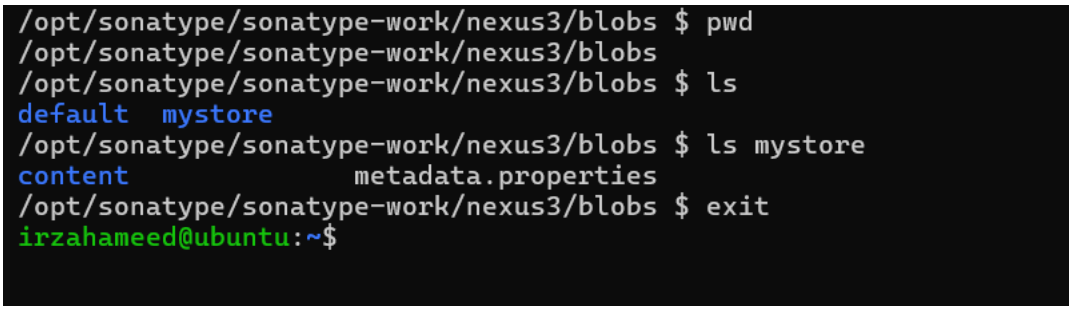
- **nexus_default_blob_content.png**



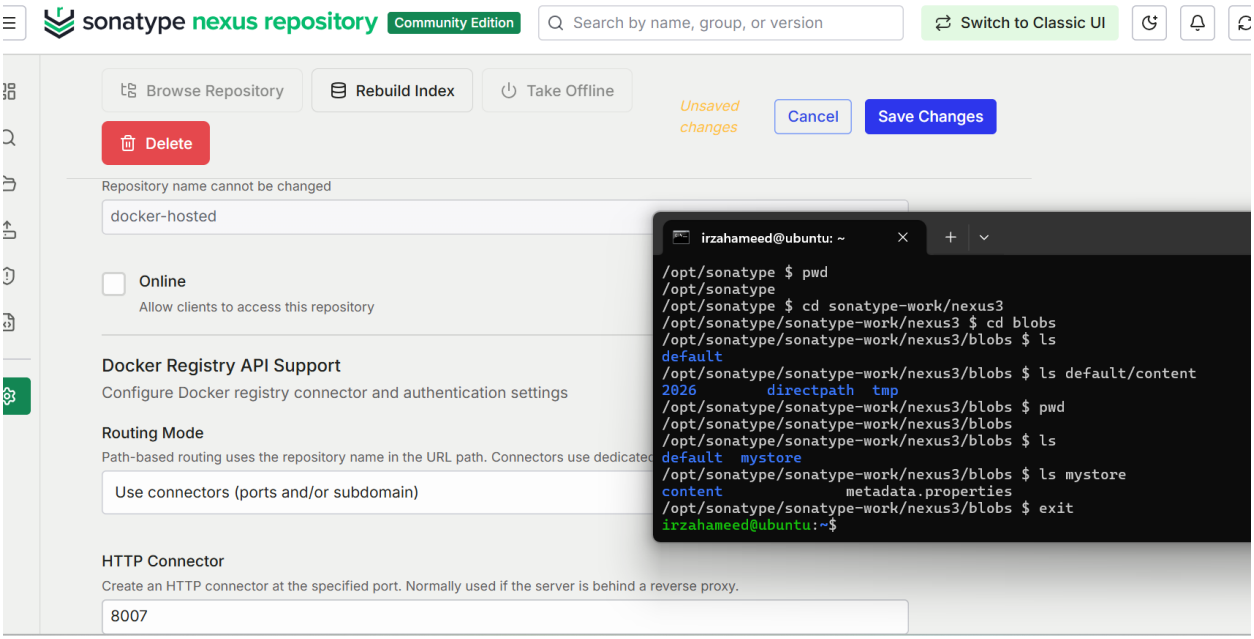
- **nexus_mystore_created.png**



- **nexus_blob_directories.png**

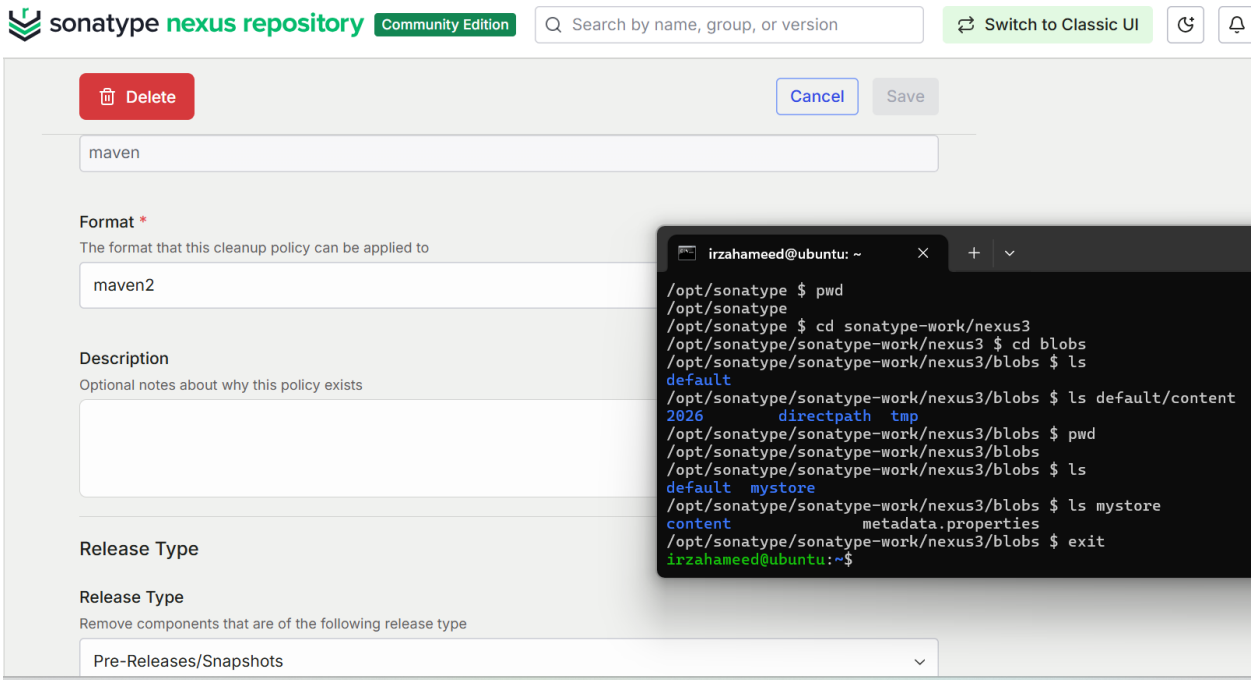


- **nexus_docker_hosted_repository.png**

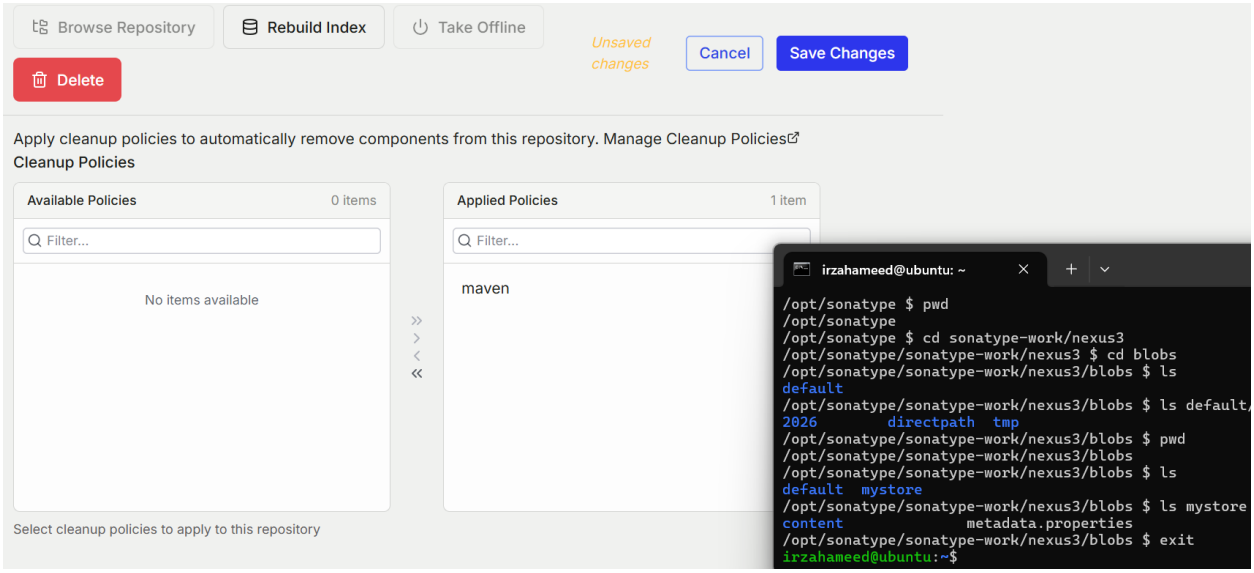


Task 7: Create and Run a Maven Snapshot Cleanup Policy

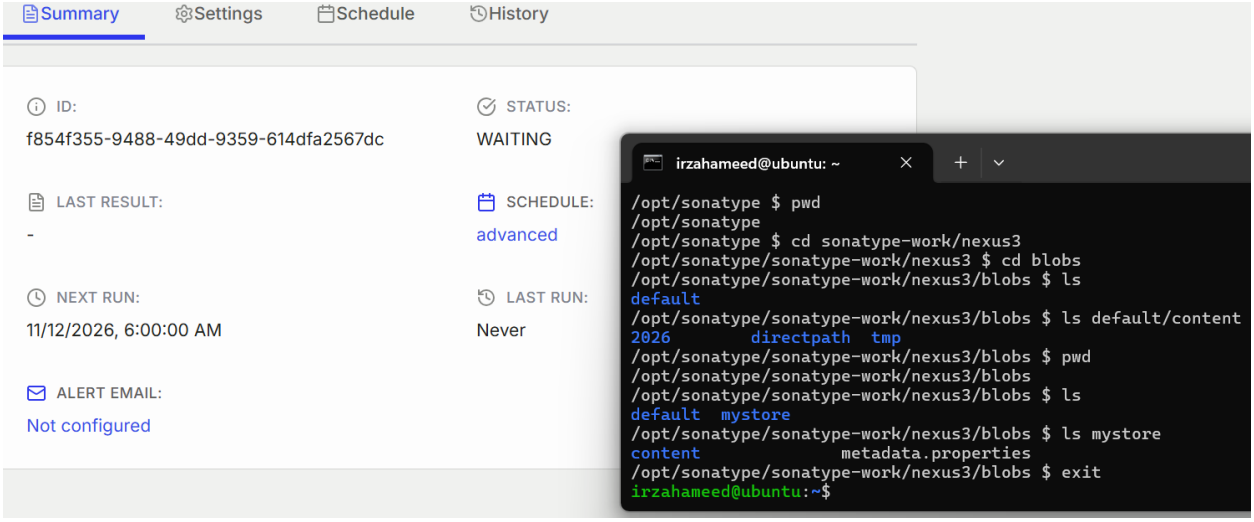
- **nexus_cleanup_policy_created.png**



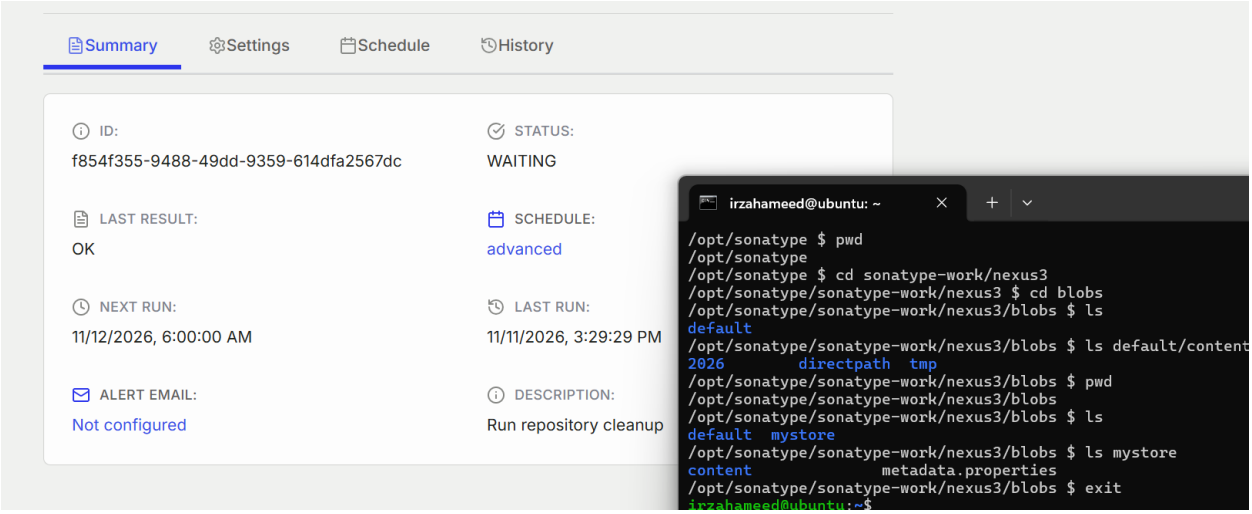
- **nexus_cleanup_policy_attached.png**



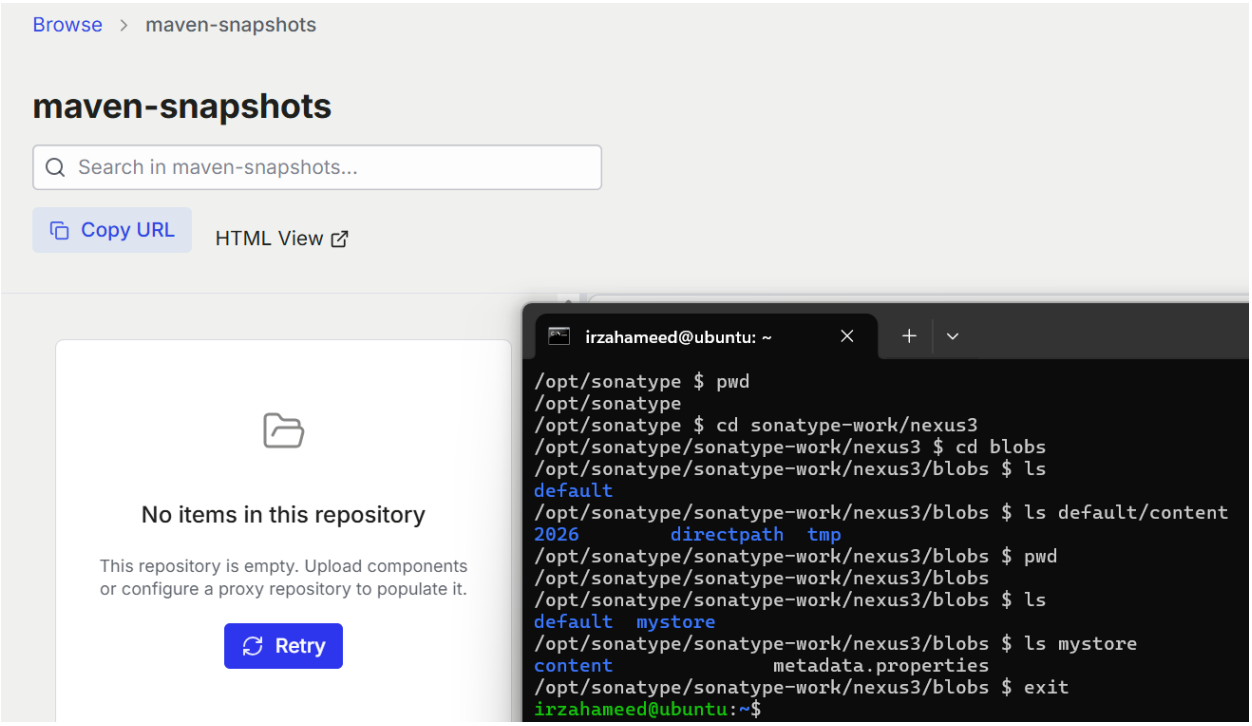
- **nexus_cleanup_task_waiting.png**



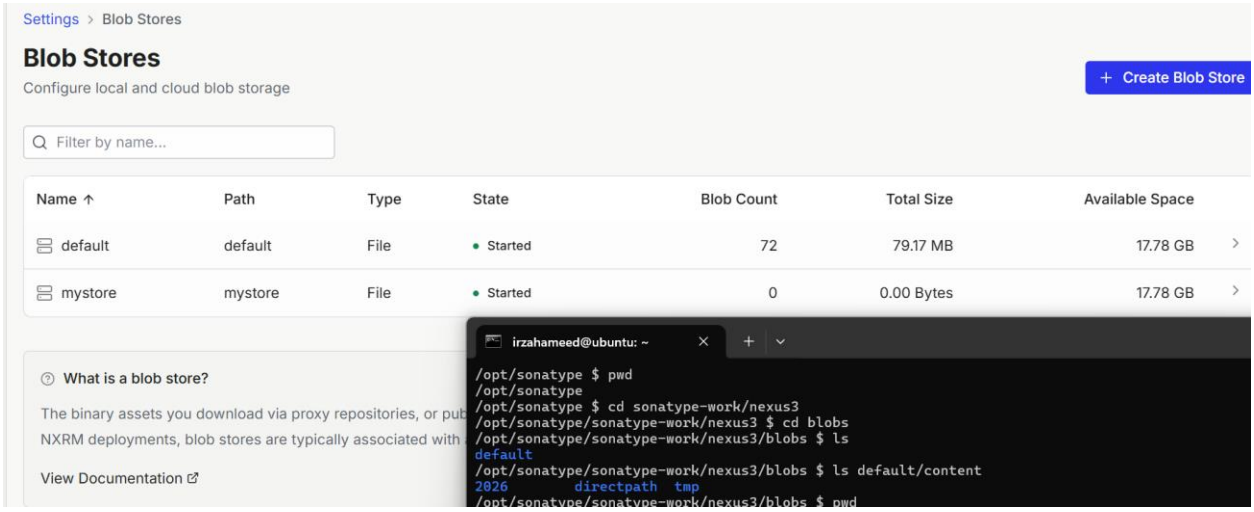
- **nexus_cleanup_task_completed.png**



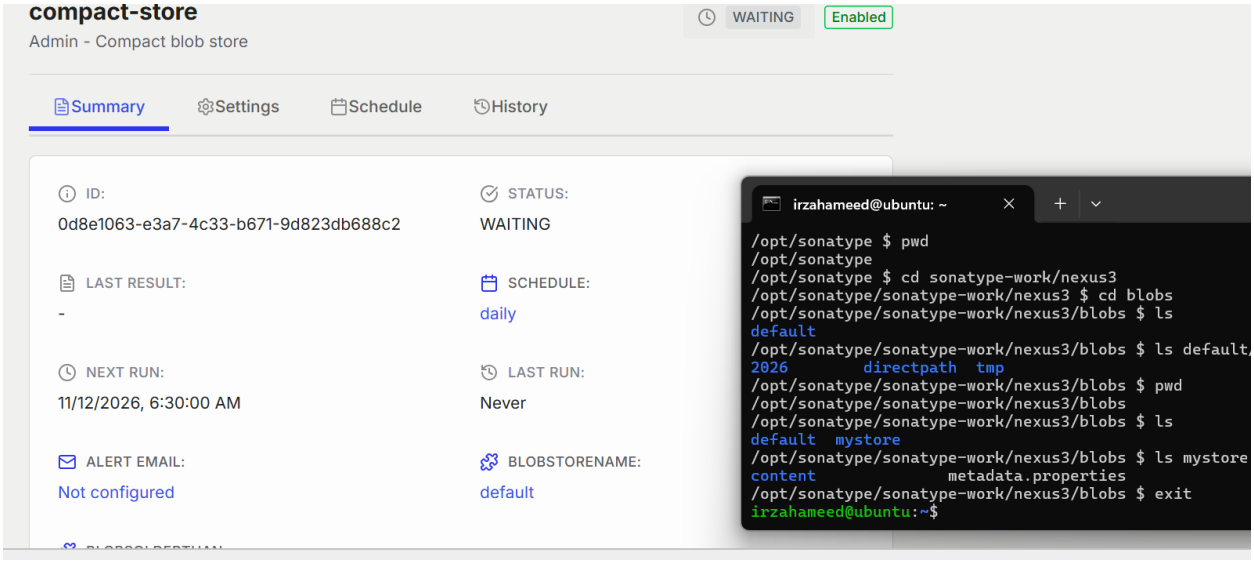
- **nexus_maven_snapshots_empty.png**



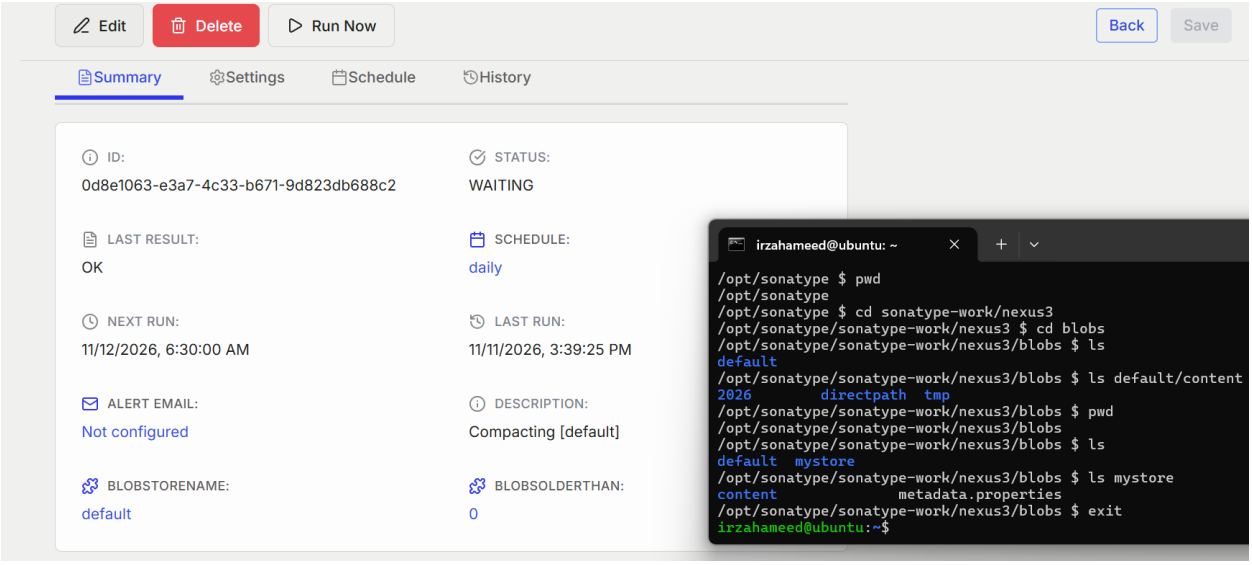
- **nexus_blob_store_before_compaction.png**



- **nexus_compact_task_configured.png**



- **nexus_compact_task_completed.png**



- **nexus_blob_store_after_compaction.png**

